

Nikhilesh Vijay Chirutala

Software Engineer

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SUMMARY

Software Engineer with 2.8+ years of experience developing and supporting production applications across cloud contact center, AI, CRM, trading, and business platforms. Strong experience with Python, FastAPI, REST APIs, AWS, Linux, Git, CI/CD, cloud application deployment, automation, monitoring, and production troubleshooting. Experienced in developing long-running Python services, supporting multi-cluster production deployments, implementing recovery workflows, and diagnosing issues across services, databases, APIs, and cloud storage. Hands-on experience with observability using Grafana and Sentry, with a strong focus on reliable deployments, operational stability, and production issue resolution.

TECHNICAL SKILLS

Languages & Scripting: Python, JavaScript (ES6+), TypeScript, SQL, Shell Scripting

Backend & Automation: FastAPI, Django, Flask, REST APIs, API Development, Backend Automation

Cloud & DevOps: AWS, Amazon S3, Linux, GitHub Actions, CI/CD, Cloud Deployment

Monitoring & Operations: Grafana, Sentry, Production Troubleshooting, Debugging, Incident Resolution, Service Monitoring

Databases: MySQL, MongoDB, PostgreSQL, SQL

Source Control: Git, GitHub, GitHub Actions

APIs & Integrations: REST APIs, Twilio, GitHub API, Postman, API Integration

Security & Authentication: Authentication, Authorization, RBAC, ACL, JWT

Frontend: React.js, Next.js, Angular, Tailwind CSS, Bootstrap, Jinja2

WORK EXPERIENCE

Python Software Engineer • CA One Tech Cloud Inc.

Apr 2026 – Present | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform

['Python', 'FastAPI', 'AWS', 'Amazon S3', 'MySQL', 'Grafana', 'Sentry']

- Develop and maintain production-grade Python services for a high-volume cloud contact center platform, supporting data processing, call-recording workflows, cloud storage, and production operations.
- Migrated a legacy Perl-based call-recording workflow to a long-running Python service deployed across 40 production clusters, supporting staged rollout, feature flags, validation, rollback, monitoring, and operational troubleshooting.
- Supported deployment and rollout of Python services across multiple production clusters, performing configuration validation, rollout verification, monitoring, and troubleshooting during migration activities.
- Developed validation and recovery workflows for production processing, handling incomplete operations, failed workflows, and inconsistent storage states to improve operational reliability.
- Built AWS S3 audit and recovery workflows for recording storage, implementing deterministic object handling and idempotent re-upload operations for incomplete or failed storage migrations.
- Implemented production observability using Grafana dashboards and Sentry to monitor service health, track errors, diagnose processing failures, and support production incident resolution.
- Investigated production issues across Python services, APIs, databases, and cloud storage, performing root-cause analysis, troubleshooting failures, and implementing fixes to maintain reliable production workflows.
- Built Python integrations with internal microservices, implementing service APIs, eligibility checks, processing controls, retries, response validation, and result persistence for distributed production workflows.

Software Engineer • Mantra Technologies

Feb 2024 – Mar 2026 | Hyderabad, India

- Worked across four production applications, contributing to backend services, APIs, application deployments, integrations, testing, debugging, and production support.
- Built backend modules, REST APIs, and business logic using Python, FastAPI, and MongoDB for application-specific production workflows.
- Developed frontend interfaces using React.js, Next.js, Angular, JavaScript, and TypeScript, integrating application workflows with backend APIs.
- Integrated third-party services including Twilio and GitHub API and supported application deployments across AWS and Vercel environments.
- Maintained GitHub Actions CI/CD workflows supporting application builds, deployment processes, and delivery of production changes.
- Performed application debugging, issue investigation, bug fixes, and production support to maintain application reliability.
- Implemented authentication, RBAC, and ACL-based authorization to enforce access controls across application resources.

Key Projects:

AI Bull — AI-Powered Trading Platform

['React.js', 'Next.js', 'FastAPI', 'Python', 'MongoDB', 'RAG', 'LLM']

- Developed an AI-powered trading and investment analysis platform for stock and mutual fund analysis, screening, and back-testing.
- Built FastAPI/Python backend APIs and business logic supporting financial data processing and application workflows.
- Developed back-testing engines and configurable screening and filtering capabilities for stocks and mutual funds.

- Implemented a Retrieval-Augmented Generation (RAG) based chatbot that retrieves relevant financial and application-specific information to generate contextual responses.
- Integrated the RAG chatbot into application workflows, enabling conversational access to relevant financial information.

CRM.io — AI-Driven CRM System

[*'FastAPI', 'Python', 'Jinja2', 'MongoDB', 'ACL', 'Twilio', 'REST APIs'*]

- Developed CRM modules for Contacts, Tasks, communication workflows, and telephony operations using FastAPI and Python.
- Implemented RBAC and ACL authorization to control access to CRM resources based on user permissions.
- Integrated Twilio for telephony and communication functionality and developed server-rendered interfaces using Jinja2.
- Tested APIs and workflows with Postman and resolved bugs across CRM functionality and production workflows.

GharPe — Real Estate Management Platform

[*'React.js', 'FastAPI', 'Python', 'MongoDB', 'REST APIs'*]

- Developed the React.js-based admin system for managing real estate projects and controlling their visibility and availability on the website.
- Built project management workflows allowing administrators to create, update, configure, and manage projects displayed across the platform.
- Developed lead management functionality to track and manage customer inquiries generated through the platform.
- Built builder management features for maintaining builder information and associating builders with their respective projects.
- Developed and integrated FastAPI/Python REST APIs supporting administrative workflows, project configuration, and data operations.

PERSONAL PROJECTS

JS Studio — Online JavaScript Playground

[*'Next.js', 'GitHub API', 'Vercel'*]

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

AnyScrum — Project Management Platform

[*'Next.js', 'Neon DB', 'Clerk', 'Vercel'*]

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

Guider — Interactive Learning Portal

[*'Angular 19', 'TypeScript', 'SSR', 'Vercel'*]

- Built an interactive learning platform using Angular 19, TypeScript, and server-side rendering with reusable frontend components.
- Deployed the application on Vercel with a production-ready frontend architecture.

Enterprise REST API Platform — Django and Flask Backend Application

[*'Python', 'Django', 'Flask', 'PostgreSQL', 'REST APIs'*]

- Built modular REST APIs using Django, Django REST Framework, and Flask for authentication, business logic, and data-driven workflows.
- Integrated PostgreSQL with CRUD operations, database models, relationships, validation, and API error handling.
- Implemented JWT authentication and role-based access control, with API testing and validation using Postman.

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur